Reg. No. | | | | | | | | | | | [17CS4IOT02]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

MICROPROCESSORS AND MICROCONTROLLERS

Time: 3 Hours **Maximum Marks:70**

Instructions:1. Answer any **FIVE** full questions. Choosing at least **TWO** questions from each Part.

2. Any missing data may be suitably assumed

PART - A

1.	**a.**	i.	Convert $1714_{(10)}$ to hexadecimal number	**6**
		ii.	Convert $9F2D_{(16)}$ to decimal number	
		iii.	Add two hexadecimal numbers 23D9+94B4	
		iv.	Subtract two hexadecimal numbers 68D-2B8	
	b.		Design full adder using half-adder.	**4**
	c.		Discuss internal organization of computer.	**4**
2.	**a.**		Sketch and explain the architecture of 8085 microprocessor.	**8**
	b.		Compare microprocessors with microcontrollers.	**6**
3.	**a.**		Discuss 8051 microcontroller with block diagram.	**8**
	b.		Explain immediate and register addressing modes.	**6**
4.	**a.**		Explain the following instruction with example	
		i. DA	ii. MUL A, B iii. DIV A, B iv. CPL A	
	b		Discuss the following pseudo-code with example:	
		i. ORG	ii. EQU iii. END	

PART - B

5.	**a.**		Explore the data types used in 8051 C-programming in detail.	**6**
	b.		Discuss different data conversion programs in 8051-C.	**8**
6.	**a.**		Write a test program for the DS89C420/30 chip to toggle all the bits of P_0, P_1 and P_2 every ¼ of a second. Assume a crystal frequency of 11.0592 MHz.	**6**
	b.		Write a 8051 ALP to do the following	**8**
		i.	When SW=0, the DAC output gives a staircase waveform.	
		ii.	When SW=1, the DAC output gives a triangular waveform.	
7.	**a.**		Explain TMOD register of 8051 timer.	**7**
	b.		Discuss the steps to be taken in programming the 8051 to transfer character bytes serially.	**7**
8.	**a.**		Explore the steps involved in 8051 when executing an interrupt.	**6**
	b.		Write a short note on the following topics:	**8**
		i.	Cross Compiler	
		ii.	Emulators	

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